

Replay a Qualified Motion Recorder

Bind one witness, one source, and one bounded session

E0 SYNTHETIC REPLAY — NO SENSOR, DISPLAY, CLOCK, OR STORAGE CLAIM

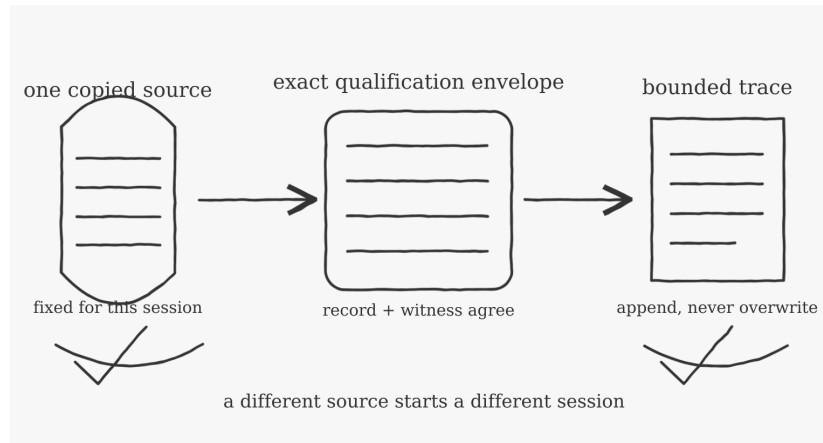
Lesson 069 — project
Prerequisites Lessons 067–068
Input copied record and qualification witness

Time 180–240 minutes
Board compile-only Mega replay
Observation named result cells

Responsibility. `QualifiedMotionRecorder` accepts one atomic copied qualification envelope, advances a versioned hand-motion script, appends a bounded caller-owned trace, and publishes inert presentation intent. It does not sample a sensor, choose between sources, drive an endpoint, read a clock, write a file, or establish durability.

1 Make interchangeability a configuration property

The same recorder contract can serve separately configured sessions. A session fixes exactly one source identity and its schema, normalization, mapping, calibration, qualification, script, and presentation revisions. Starting with one source and quietly accepting another is not interchangeability; it is an unproved source switch.



Alternative text: a rough graphite source card points through an exact qualification envelope into a bounded notebook; captions state that one source is fixed per session, record and witness agree, and appends never overwrite.

At E0, the positive replay uses synthetic copied fixtures. MPU6050-shaped and QMI8658-shaped records can exercise negative seams, but an enum tag does not identify silicon, a carrier, voltage limits, address straps, scaling, mounting, or axis direction. An unidentified board remains unpowered.

E0 establishes	E0 does not establish
one configured source per session	discovery, voting, failover, or hot swap
exact copied evidence correlation	physical source identity or acquisition
bounded volatile trace semantics	RTC time, filesystem, SD write, or durability
inert orientation and display intent	powered LCD, RGB LED, or button behavior

2 Accept one envelope, not two snapshots

The input binds the complete Lesson 067 normalized record to the complete Lesson 068 qualification witness for that exact record. Source identity, revisions, sample sequence, observation time, qualification attempt, state, reason, and evidence digest must agree. A digest helps detect a collision; it never replaces comparison of the authoritative fields.

Reading one mutable record and then a separately mutable decision could join evidence from different instants. The recorder therefore validates one copied envelope before it changes any session state.

Envelope	Recorder action	State change
qualified and exactly bound	derive, stage, append	one atomic commit
collecting or rejected	publish explicit health	no append or script advance
record/witness mismatch	reject as correlation fault	none
changed duplicate	reject as chronology fault	none
byte-identical duplicate	idempotent replay	none

Suppose record sequence 42 is paired with a witness for sequence 41. Should a plausible acceleration vector be appended? _____. Which field first proves the mismatch? _____.

3 Append without silent loss

The caller owns fixed-capacity trace storage. A qualified update stages a complete canonical cell, validates it, writes it to the next free slot, and then commits count, recorder sequence, script progress, and result together. If validation or capacity fails, the prior policy, output canaries, and every trace byte remain unchanged.

The trace never wraps, overwrites its oldest cell, silently coalesces samples, or reports a dropped sample as recorded. The last free slot may be filled; the next otherwise valid update reports **TraceFull**. Record-sequence exhaustion faults before it can wrap to zero.

Volatile boundary. Here “recorded” means copied into caller-owned memory during this process lifetime. It does not mean that media acknowledged a write, a filesystem committed, a clock was accurate, or evidence survives reset or power loss.

4 Give the trace a normative image

Record identity is an explicit fixed-width encoding with magic, format and length, source and revision identity, session and recorder sequences, observation time, qualification attempt, script step, normalized values, quality, orientation, health, reserved-zero fields, and an integrity check. Field order, widths, signed representation, and byte order are part of that contract. C++ padding, an address, or `memcmp` of public objects is never record identity.

1. Corrupt one reserved byte. Predict the decode result: _____.
2. Replay a byte-identical accepted envelope. New trace count: _____.
3. Change one vector component while retaining its sequence. Outcome: _____.

5 Advance a script only on accepted evidence

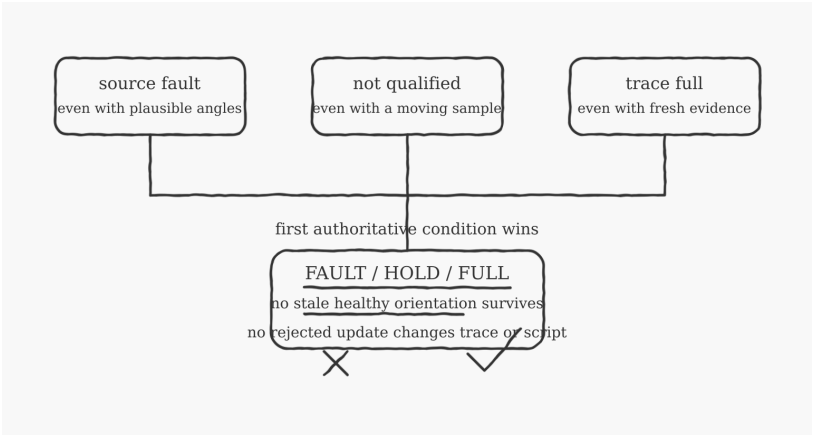
Each accepted record belongs to the current versioned hand-motion step before the completion predicate advances the script. Rejected, stale, saturated, unqualified, or duplicate evidence cannot advance it. The terminal step stays terminal until reset.

Lesson 044 remains the only pitch/roll presentation algorithm. Lesson 069 feeds it the qualification-frame sample and copies its result into intent; it does not invent another orientation convention. Record sequence, source sequence, qualification attempt, session generation, and script step remain different values even when a fixture makes their numbers happen to match.

Event	Trace	Script
qualified step incomplete	append current step	hold
qualified step complete	append current step	advance afterward
unqualified plausible motion	unchanged	hold
terminal-step replay	append only if new and accepted	remain terminal
reset	clear declared volatile session state	begin at step zero

6 Let faults dominate presentation

Presentation is copied intent: source and recorder health, script prompt and progress, orientation value-or-fault, trace count and full indication, character-display tokens, and RGB color/blink tokens. It owns no pins and causes no electrical action.



Alternative text: rough graphite cards for source fault, not qualified, and trace full converge on one fault-dominant presentation card; a crossed-out stale angle and check-marked unchanged trace show that rejected updates neither look healthy nor alter stored evidence.

Structural/configuration failure precedes lifecycle/correlation failure, which precedes producer fault, source-quality fault, age/order fault, qualification failure, and healthy operation. Lower-priority evidence remains inspectable where the result provides a field, but it cannot repaint a higher-priority fault as a healthy angle.

Collision	Required interpretation
source fault plus plausible orientation	source fault; no current angle
qualification rejection plus apparent step motion	rejected; no advance
full trace plus fresh qualified record	full; no overwrite
shutdown plus pending presentation	canonical inert presentation

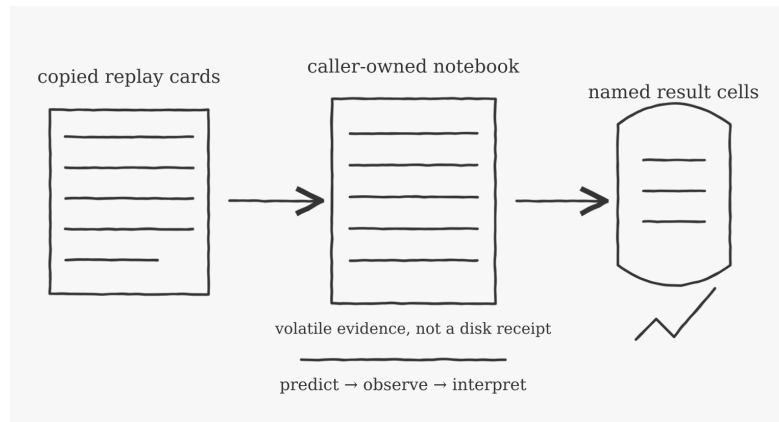
7 Treat modular ordering deliberately

Supplied time is modular **TimePoint**, never silently UTC. Equal sequence/time is idempotent only for byte-identical evidence. Regression, future observation, changed duplicates, exact half-range ambiguity, age beyond the configured inclusive boundary, and exhaustion reject atomically.

At the exact maximum-age boundary evidence is _____; one representable tick later it is _____. Explain why a signed cast is not a proof of modular order: _____.

8 Replay without touching hardware

The canonical Mega sketch constructs copied records and qualification witnesses, supplies time, invokes the recorder, and copies health, script, orientation validity, trace count, capacity state, RGB intent, and display intent into named volatile result cells.



Alternative text: rough graphite replay cards point into a caller-owned lined notebook and then named result cells; a caption identifies the notebook as volatile evidence rather than a disk receipt.

1. **Predict:** write the accepted source, health, current script step, orientation validity, trace count, and presentation tokens.
2. **Observe:** compile for Mega 2560 and inspect the named volatile cells in a debugger, simulator, or disassembly. Serial may support the observation but is not required.
3. **Interpret:** compare the complete copied envelope, resulting trace cell, script attribution, count, and fault intent. Explain any disagreement before changing the fixture.

This is the E0 non-Serial observation path. It proves deterministic software publication separately from electrical acquisition and safe state. E0 owns no endpoint, so physical resource-acquisition and released-line evidence remain open.

9 Publisher verification and open acceptance

Host evidence	Strict lifecycle, envelope-correlation, capacity, canary, ordering, encoding, script, and failure-collision tests pass.
Mega evidence	The synthetic replay compiles and linked flash/static SRAM, stack, object, result, image, caller trace, and residual SRAM are measured.
Document evidence	PDF metadata, extraction, page count, monochrome, visual classification, and rendered-pencil review pass.
Physical evidence	Open: exact sensor/carrier, mounting, bus, display/RGB, button, RTC, SD/media, clock, persistence, and durability acceptance.

No physical verification may be inferred from a host test or a compiled sketch. Future powered work must separately record prediction, where and when to observe a non-Serial result, interpretation, resource-acquisition evidence, and safe-state evidence.

10 Stress the complete project

Exercise capacity one, one-before-full, full, unchanged duplicate, changed duplicate, ordinary rollover, exact half-range ambiguity, source fault, qualification rejection, stale evidence, reset, repeated shutdown, and reinitialize. For each rejection, prove both the recorder state and every caller-owned trace byte are unchanged.

The maximum composition includes Lessons 043, 044, 067, 068, and 069, the largest qualification window, staged cell, complete envelope, orientation result, presentation result, and maximum caller-owned trace counted exactly once. It performs bounded work per update: no polling, acquisition wait, display refresh loop, media retry, or unbounded catch-up.

11 Draw the powered boundary

There is no wiring diagram or formal schematic at E0. A future implementation must split acceptance into independently qualified work:

- exact sensor chip/carrier, I2C identity, voltage, pull-ups, scaling, data-ready behavior, mounting, and calibration;
- exact display, RGB LED, and button ownership, polarity, current, startup, self-test, fault dominance, and safe state;
- exact RTC, SD/media, timestamp domain, transaction receipts, schema, corruption and torn-write recovery, capacity, wear, removal, and power-loss durability.

This lesson authorizes only hand motion of an inert identified board. It authorizes no motor, vehicle control, launcher, ignition, stored-energy release, or unattended moving load.

12 Exercises and acceptance reasoning

1. Run two separate sessions with different copied source identities. Show why neither session contains a midstream substitution.
2. Pair a fresh record with an older qualified witness. List every observable byte and state value that must remain unchanged.
3. Fill the last trace cell, then offer one fresh qualified envelope. Distinguish “full” from source fault and from durable storage.
4. Inject source fault while retaining a plausible cached angle. Explain the required orientation and RGB/display intent.
5. Corrupt each canonical image field class: framing, enum, reserved, payload, and integrity. Confirm decoding never partly publishes.
6. List the physical evidence needed before replacing copied fixtures with one exact sensor, display, clock, and media assembly.

Project result. You have composed normalized and qualified copied motion evidence into a one-source, bounded, fault-dominant recorder replay without smuggling source selection, hardware control, wall-clock meaning, or durable storage into E0.

Stop at E0: every powered specimen, wiring, mounting, presentation, clock, media, durability, and physical acceptance result remains open work.